BiasAperture — Research Sprint: Executive Synthesis

Target Audience:	Project Leads, Fellowship Supervisors, Technical Assessors, Viva Defense Panel
Document Level:	High-Level (Executive, Novelty Defense, Regulatory Mapping & Strategic Overview)
Date:	August 2026
Context:	Milestone M1 Completion & 20-Track Parallel Research Sprint

1 Executive Summary & Vision

BiasAperture is an open-source, strictly diagnostic demographic bias auditing platform for computer vision (facial analysis) models.

Modern facial analysis systems exhibit well-documented performance disparities across intersectional demographic groups (e.g., race, gender, and age). While foundational academic toolkits exist for tabular algorithmic fairness (such as Fairlearn and AIF360), they remain disconnected from the realities of computer vision pipelines, lack standardized statistical safeguards against sample-size artifacts, and do not bridge the gap between technical metric outputs and emerging regulatory mandates.

BiasAperture bridges this operational gap. It consumes facial image datasets and model predictions, executes dual-backend cross-validated fairness computations, enforces hard sample-size ($n \geq 30$) and uncertainty (bootstrap CI) guards, triggers targeted local explainability (SHAP) on statistically significant disparities, and exports a single-file, regulator-legible compliance report aligned with the **EU Artificial Intelligence Act (Regulation EU 2024/1689)** and the **NIST AI Risk Management Framework (AI 100-1)**.

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                    BIASAPERTURE ARCHITECTURAL FLOW

[ Computer Vision Model & Dataset ]
                 |
                 v
  [ 1. Data & Model Ingestion ] -------> dlib CNN Align + Schema
                 |                        Invariant Validation
                 v
  [ 2. Fairness Detection Engine ] -----> Dual-Backend Cross-Validation
                 |                        (Fairlearn + AIF360)
                 |                        - Core Four: DPD, EOD, EOP, DIR
                 |                        - Chi-Squared (p<0.05) + 95% BCa CI
                 v
  [ 3. Targeted Explainability ] -------> Triggered on Flagged Disparities
                 |                        (p < alpha, n >= 30)
                 |                        - SHAP (Partition / Gradient)
                 |                        - Proxy Detection (Attribution
                 |                          Shift + ITA Skin Tone)
                 v
  [ 4. Compliance Report Generator ] ---> Self-Contained HTML
                                           (Mitchell Model Card +
                                            Gebru Datasheet)
                                           Regulatory Tracing: EU AI
                                           Act Art. 10 & NIST AI RMF
```

2 The Five Operating Modules

BiasAperture is structured into five cohesive work packages (WPs) operating under strict scope boundaries:

1. **WP1: Baseline & Schema Lock (Completed)**
   - Locked internal schema contracts (`SubjectRecord`, `MetricResult`) and taxonomies (7 race categories, 2 gender categories, 9 age bins).
   - Fixed the baseline reference classifier (`dchen236/FairFace` ResNet-34 multi-task architecture).
2. **WP2: Data Ingestion & Model Interfacing (Stream A)**
   - Dual-mode ingestion: `PredictionsFileInterface` (non-negotiable core batch CSV/JSON ingestion) and `InProcessInterface` (live PyTorch inference).
   - Two-mode validation (strict/fail-fast vs. profiling) verifying schema conformance and catching missing/corrupt values.
3. **WP3: Compliance Report Generation (Stream B)**
   - Single-file standalone HTML report generation using Jinja2 with inline CSS and base64-encoded visual artifacts.
   - Built-in **Model Cards for Model Reporting** (Mitchell et al., 2019, 9 sections) and **Datasheets for Datasets** (Gebru et al., 2018).
4. **WP4: Statistical Fairness Engine & Explainability (WP4 / Streams C, D, E)**
   - Evaluates the **Core Four** fairness metrics: Demographic Parity Difference (DPD), Equalized Odds Difference (EOD), Equal Opportunity Difference (EOP), and Disparate

Impact Ratio (DIR).

- Enforces the **NFR-003 sample-size guard** ($n \geq 30$) to eliminate false-positive disparities caused by small-cell noise.
- Computes exact χ^2 independence tests (p-value, $\alpha = 0.05$) and 95% Bootstrap Confidence Intervals ($B \geq 1{,}000$ resamples).
- Targeted local SHAP visual attribution and proxy variable detection.

5. **WP5: System Orchestration & CLI (Integration)**
   - High-level facade orchestrating the pipeline from raw ingestion to the final exported compliance dossier.

3 The 20-Track Research Sprint: Key Findings

To completely de-risk implementation, a comprehensive 20-track parallel research sprint was executed across 6 thematic streams:

Stream	Focus Area	Tracks	Critical Takeaways & Decisions
Stream A	Data Ingestion	01–04	FairFace uses a ResNet-34 backbone with a single 18-unit multi-task head and `dlib` 5-point alignment (not MTCNN). Released dataset on disk contains **97,698 images** (86,744 train, 10,954 val). **UTKFace is formally cut** due to 3/7 race-mapping collapses and DEX age noise.
Stream B	Report Gen	05–08	Single-file HTML generator with zero CDN/JS runtime dependencies. Custom Jinja2 implementation chosen over `model-card-toolkit` (avoids rigid TFX dependencies). Complete sub-clause mapping for EU AI Act Article 10(2)–10(5).
Stream C	Fairness Math	09–14	Discovered mathematical divergence between Fairlearn (max-of-gaps) and AIF360 (mean-of-gaps) for EOD; unified under max-of-gaps. Uncovered that AIF360 returns signed EOP while Fairlearn is unsigned (fixed with `abs()`). Proved $3\times$ disparity skew if $n < 30$ groups are filtered post-computation rather than pre-computation. Designed custom BCa/percentile bootstrap because `scipy.stats.bootstrap` fails on multi-group metrics.
Stream D	Explainability	15–16	Selected `PartitionExplainer` as black-box default and `GradientExplainer` for live PyTorch models. Designed dual-signal proxy detection combining spatial SHAP shift with Individual Typology Angle (ITA) skin-tone colorimetry. Grounded limitations in Bilodeau et al. (2022) impossibility results.

Stream	Focus Area	Tracks	Critical Takeaways & Decisions
Stream E	Architecture	17–18	Formulated `FairnessBackend` Strategy pattern and `CrossValidationOrchestrator`. Built shared sample-size logic in `base.py` to prevent false backend divergence. Engineered known-answer 8-record test suite and Hypothesis property tests.
Stream F	Defense	19–20	Mapped platform directly to NIST AI RMF **Measure** function (2.11, 2.3, 1.1). Executed competitive audit against 7 industry tools confirming unique positioning.

4 Defensible Novelty & Competitor Positioning

A common question during technical defenses is: *"Why build a new tool when Fairlearn, AIF360, and Aequitas already exist?"*

The novelty of BiasAperture lies not in reinventing individual mathematical equations, but in **solving workflow friction through end-to-end architectural integration**. Existing tools suffer from severe domain fragmentation:

Tool / Framework	CV & 7-Race Ingestion	Dual-Backend Cross-Val	Stat Guards ($n \geq 30$+CI)	Regulatory Mapping	Targeted Local SHAP
Aequitas	N	N	N	N	N (global plots only)
Fairlearn	N (tabular)	N	N	N	N
AIF360	N (tabular)	N	N	N	N
Google What-If	N (interactive)	N	N	N	N
JFAM (Alg. Audit)	N (unlabeled)	N	partial	N	N
FAT Forensics	N	N	N	N	partial (own module)
BiasAperture	**Y (native)**	**Y (locked)**	**Y (hard)**	**Y (EU/NIST)**	**Y (targeted post)**

The 5 Unique Architectural Differentiators

1. **CV-Native Demographic Pipeline**: Ingests raw multi-output predictions and aligns them directly into a 7-race, 2-gender, 9-age schema without manual tabular wrangling.

2. **Dual-Backend Cross-Validation**: Executes computations across two independent engines (Fairlearn & AIF360) to catch implementation artifacts, API bugs, and numerical edge cases.

3. **Hard Statistical Invariants**: Enforces $n \geq 30$ sample size guards at the dataclass level (`MetricResult.__post_init__`) and calculates 95% BCa Bootstrap CIs with χ^2 significance for every row.

4. **Targeted Explainability**: Automatically triggers SHAP and proxy variable detection *only* on rows that exhibit statistically verified bias ($p < 0.05$, $n \geq 30$), saving massive computational overhead.

5. **Direct Regulatory Provenance**: Directly maps each technical finding to EU AI Act Article 10 sub-clauses and NIST AI RMF Measure subcategories.

5 Regulatory Alignment: EU AI Act & NIST AI RMF

EU Artificial Intelligence Act (Regulation EU 2024/1689)

BiasAperture serves as a technical verification mechanism for high-risk AI systems:

- **Article 10(2)(f)**: Mandates examination of potential biases in training/validation data. *Satisfied by the Core Four metric evaluations.*
- **Article 10(2)(g)**: Requires measures to detect, prevent, and mitigate bias. *Satisfies the detection requirement (mitigation is explicitly out of scope).*
- **Article 10(3)**: Mandates that datasets have appropriate statistical properties. *Directly satisfied by the* $n \geq 30$ *sample size guard and 95% bootstrap confidence bounds.*
- **Article 10(5)**: Exceptional processing of special-category demographic data for bias detection. *Documented within the report's governance metadata.*
- **Annex IV §2(g)**: Technical documentation requirements for validation and test procedures.

NIST AI Risk Management Framework (NIST AI 100-1)

BiasAperture instruments the **Measure** core function:

- **Measure 2.11**: Fairness and bias are evaluated, quantified, and documented.
- **Measure 1.1**: Risks that cannot be reliably quantified are documented rather than ignored (*the exact structural role of* `insufficient_sample=True, metric_value=None`).
- **Measure 1.3**: Rigorous assessment through independent verification mechanisms (*realized via dual-backend cross-validation*).
- **Measure 2.9**: Explainability and interpretability metrics (*realized via targeted SHAP pixel attributions*).

6 Scope Invariants & Descoping Strategy (Cut-List)

To guarantee deterministic, high-quality completion on the fellowship schedule, the project operates under strict non-negotiable boundaries and an established cut-list.

What BiasAperture Will NEVER Do

- **No Model Retraining / Fine-Tuning**: BiasAperture is strictly diagnostic.
- **No In-Processing Weight Debiasing**: Does not alter weights or loss functions.
- **No Synthetic Data Generation**: Does not generate synthetic demographic faces.

Formal Cut-List (Ordered by Drop Priority)

1. **Web UI**: Drop Streamlit/Flask; retain robust CLI and static HTML exports.
2. **UTKFace Benchmark**: Formally dropped to focus 100% of engineering bandwidth on FairFace.

3. **PDF Generation**: Retain standalone HTML; drop headless browser PDF compilation.
4. **In-Process PyTorch Inference**: Fall back strictly to batch predictions file ingestion (`PredictionsFileInterface`).
5. **AIF360 Backend**: Retain Fairlearn as standalone engine (only as an emergency measure).

7 Current Project State & Next Steps

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Milestone Progress:
[========------------] 40%
+-- WP1: Schema Lock & Baseline           (100% - LOCKED)
+-- 20-Track Research Sprint              (100% - SYNTHESIZED)
+-- WP2: Data Ingestion & Test Matrix     (In Progress - 40%)
+-- WP3: Compliance Reporting             (In Progress - 30%)
+-- WP4: Fairness Engine & Explainability (In Progress - 20%)
+-- WP5: Integration & CLI                (Scheduled - 0%)
```

With the research sprint complete, the implementation path is clear:

1. Build `bias_aperture/data_ingestion.py` and finalize the test matrix.
2. Implement `bias_aperture/report/generator.py` and the Jinja2 HTML templates.
3. Implement `bias_aperture/fairness/base.py`, backend adapters, bootstrap statistics, and SHAP explainability.
4. Wire the end-to-end CLI orchestrator in WP5.